

Hanyoung Jung

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Bachelor of Science in Statistics and Machine Learning

Aug. 2024 – May. 2028

- Relevant Coursework: Discrete Mathematics & Logic, Data Structures & Algorithms (C), Probability & Statistical Inference I–II, Introduction to Machine Learning

EXPERIENCE

Undergraduate Researcher

May 2026 – Present

MetaMobility Lab, Carnegie Mellon University

Pittsburgh, PA

- Developed TCN-based models for perturbation detection and center-of-mass (CoM) velocity estimation using wearable exoskeleton sensor data.
- Built a motion-capture data pipeline to derive CoM velocity ground truth across 5 subjects for training and evaluating TCN regression models.
- Optimized TCN inference for Teensy deployment using INT8 quantization and cached overlapping convolution computations, accelerating inference from 1.5 to 130–350 FPS to reduce reliance on Jetson GPU.

Machine Learning Research Intern

May 2025 – Aug 2025

EXEM

Seoul, Korea

- Implemented log anomaly detection models (DeepLog, LogBERT, ensemble/meta-models), achieving 99.7% accuracy and 0.95+ F1 on 16K+ production logs.
- Built a multilingual RAG-based LLM system (Korean, English, Japanese, Chinese) using Qdrant vector search, semantic retrieval, FastAPI backend, and React.js frontend framework.
- Presented technical talks on embeddings, gradient descent, and modern ML model architectures to internal engineering teams.

Neuroscience Research Intern

July 2022 – Aug 2022

Korean Basic Science Institute

Seoul, Korea

- Collected and processed experimental neuroscience data from mouse brain tissue experiments and mass spectrometry analyses.
- Analyzed experimental results to identify potential depression-related biomarkers and biological patterns in molecular data.

PROJECTS

Multilingual RAG LLM System | *Python, FastAPI, Qdrant, LLMs*

July 2025 – Aug 2025

- Crawled and processed tourism website content from Gangwon Province to construct the retrieval corpus.
- Built a multilingual retrieval-augmented generation system for Gangwon Province tourism information, supporting Korean, English, Japanese, and Chinese queries.
- Implemented semantic retrieval using Qdrant vector search and embedding-based similarity matching.
- Developed a FastAPI backend and React frontend for an interactive multilingual QA interface.

HONORS

AMC 12A Honor Roll

2023

Top 5% of contestants

USA Computing Olympiad Silver Division

2023

Top 15% of contestants

TECHNICAL SKILLS

Computer Languages: Python, C/C++, SQL (Postgres), JavaScript, R

ML / Data: PyTorch, NumPy, pandas, scikit-learn

Systems / Tools: FastAPI, Docker, Git, Qdrant

Languages: English, Korean, Japanese